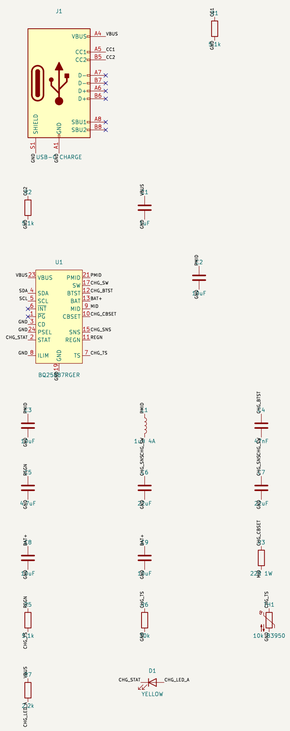
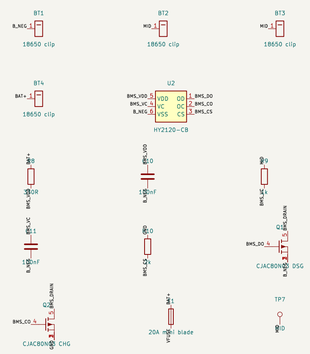
1 USB-C charge input + BQ25887 2S balance charger



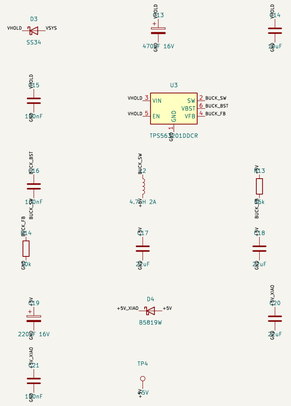
2 Cells, 2S BMS (HY2120), 20 A blade fuse



3 Power switch (P-FET), surge clamp, bulk



4 Hold-up + 5 V buck (TPS563201), XIAO feed



5 XIAO ESP32-C3, I2C

6 DRV8833 drive, 1.3 A current limit

7 PWM outputs (ESC or servo) + spare GPIO

8 RGB status LED

Power flags (ERC)

#FLG01 PWR_FLAG
#FLG02 PWR_FLAG
#FLG03 PWR_FLAG
#FLG04 PWR_FLAG
#FLG05 PWR_FLAG
#FLG06 PWR_FLAG
#FLG07 PWR_FLAG
#FLG08 PWR_FLAG
#FLG09 PWR_FLAG
#FLG010 PWR_FLAG
#FLG011 PWR_FLAG
#FLG012 PWR_FLAG
#FLG013 PWR_FLAG
#FLG014 PWR_FLAG
#FLG015 PWR_FLAG

Cardboard Bot v2 main board — rev A — Columbia Gadget Works
Generated from hardware/gen/circuit.py (do not hand-edit; edit circuit.py and regenerate).
All connections are made by net labels at pin ends.
XIAO pins: D0 SDA(GPIO2) D1 IO3 D2–D5 DRV8833 AIN1/AIN2/BIN1/BIN2 D6 IO21 D7 OUT2(GPIO20) D8 SCL(GPIO8) D9 LED(GPIO9) D10 OUT1(GPIO10)
Values to confirm against datasheets before ordering: BQ25887 caps (SNS 44uF, PMID, REGN, BTST), L1, CBSET R3, TS divider, HY2120 RC values.

Columbia Gadget Works	
Sheet: /	
File: cardboard-bot.kicad_sch	
Title: Cardboard Bot v2 main board	
Size: A1	Date: 2026-10-27 — **Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 1/1